

AI Meeting Memory Vault using MongoDB (Console-Based REST API Project)

Objective

The AI Meeting Memory Vault is a console-based backend application designed to store, organize, and retrieve meeting information efficiently. The system allows users to upload meeting transcripts, generate and save AI-powered summaries, assign tags to meetings, and search historical meeting records. The project demonstrates CRUD operations, REST API development, MongoDB document relationships, text indexing, full-text search, embedded documents, and aggregation pipelines.

Technologies Used

- Runtime Environment: Python
- Backend Framework: FastAPI
- Database: MongoDB
- API Testing Tools: Swagger UI / Postman
- Version Control: Git & GitHub
- Development Type: Console-Based Backend Application

System Entities

1. Users
2. Meetings
3. Meeting Transcripts
4. AI Summaries
5. Tags

Working of the System

1. Users create and manage meeting records through REST APIs.
2. Meeting transcripts are uploaded and stored in MongoDB.
3. AI-generated summaries are created and linked to their respective meetings.
4. Users can assign multiple tags to categorize meetings.
5. All meeting information is stored as MongoDB documents with embedded metadata.
6. Users can retrieve, update, or delete meeting records using API endpoints.
7. Full-text search allows users to find meetings using keywords from transcripts or summaries.
8. Tags help filter meetings based on project, department, or topic.
9. MongoDB Aggregation Pipelines generate analytical insights from stored meeting data.
10. The system provides quick access to historical discussions and decisions.

MongoDB Operations Implemented

- Create Meeting
- View Meeting
- Update Meeting
- Delete Meeting
- Upload Transcript
- Save AI Summary
- Add Tags
- Search Meetings
- Filter Meetings
- Generate Analytics
- Aggregation Queries
- Text Search Indexing
- Complete CRUD Operations

MongoDB Relationships

- One User → Many Meetings
- One Meeting → One AI Summary
- One Meeting → Many Tags
- One User → Many Meeting Records

REST API Workflow

User Request → FastAPI REST API → Business Logic Layer → MongoDB Database → Aggregation & Search Processing → API Response → Console/Postman Output

Expected Outcome

The system provides a centralized repository for storing and retrieving meeting knowledge. By utilizing MongoDB's flexible document model, text indexing, full-text search, embedded documents, and aggregation framework, users can efficiently manage meeting records, access AI-generated summaries, and analyze historical meeting information for better decision-making.